

CHE 321: Chemical Reaction Engineering

Fall 2019 Course Syllabus

Instructor

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Office Hours – M 1:15 – 2:15 pm

Teaching Assistant and Office Hours

TBA

Office Hours: TBD

Course Email Address

All the questions, and concerns related to the course must be sent directly to the teaching assistants (TAs) or the instructor with the subject line starting with [CHE 321]

Lectures

Mondays and Wednesdays, 12:00 – 1:15 PM in Engineering Innovation Building, 124

Course Description

Chemical reaction engineering, almost exclusively the domain of chemical engineers, has been an area of particularly vigorous research activity. It comprises a complex medley of different disciplines and there is little wonder that widely varying treatments are available of the subject. Broadly, however, the behavior of chemical reactors is intimately related to the interplay of physical and chemical rate processes in various reactor settings. The investigation of chemical reactors involves a *synthesis* of information pertaining to the chemical reacting system such as kinetics (reaction rate expressions), thermodynamics (equilibrium compositions, heat of reaction etc.) and those relating to pertinent physical rate processes (momentum, heat and mass transfer). The basic conservation principles (conservation of mass, momentum and energy) appended by appropriate phenomenological descriptions of physical processes (e.g., Fourier's law of heat conduction) and economic considerations will govern suitable mathematical models often admixed with some empiricism and by profuse applications of relevant mathematical techniques.

A unified treatment of chemical reaction engineering is complicated by at least two sources of divergent behavior. Naturally, the diversity of chemically reacting systems would be inherited by the reactors in which reactions are carried out. Furthermore, reaction equipment in themselves cover a wide variety such as stirred tank, packed bed, moving bed, fluidized bed, trickle bed and other types of reactors in which the relative role of physical and chemical rate processes may be profoundly different. In the face of such diversity, it should be evident that *general* results on reactor behavior or design prescriptions are not to be found. However, the divergent features of chemical reaction

engineering have been presented in a proper perspective by Aris. The student is urged to develop an appreciation for this "morphological view" of the subject to understand clearly the roles and relationships of the different forms of activity in reaction engineering.

There has been an increasing number of books on reaction engineering analysis in recent times with widely varying directions of emphasis. This course will use my Class Notes in which we will refer to research articles and various sections in the following books:

1. ***Chemical Reactor Analysis and Design***, by G. Froment, K. B. Bischoff, Juray De Wilde, 3rd Edition: John Wiley & Sons, 2010. **(FBD)**: *This book presents a wide range of topics in reactor analysis. While primary reliance is made on my lecture notes, I will try to relate lecture material to reading assignments from appropriate sections in the text book FBD.*
2. ***Chemical Reactor Theory. A Review***, Edited by L. Lapidus and N. R. Amundson, Prentice-Hall, 1977. **(LA)**: *This contains several articles on various aspects of reaction engineering.*
3. ***Elements of Chemical Reaction Engineering***, by H. Scott Fogler, Prentice-Hall, 1986 **(HSF)** is sensitive to applications such as chemical vapor deposition.

The following outline of the course is tentative but generally represents the direction of emphasis. Numbers in parenthesis refer to related sections in the books mentioned above. Other references will be provided during the progress of the course.

Course Outline

1. *1st Week*
Review of High school Chemistry: Chemical reactions, thermodynamics and kinetics (Class Notes).
2. *2nd Week*
Kinematics of chemical reactions. Stoichiometry and the algebraic implications of the law of mass action. (Class Notes; **FBD** 1.1, 1.2). The intrinsic reaction rate for homogeneous and heterogeneous chemical reactions.
3. *3rd Week*
The conservation equations for chemically reacting multicomponent mixtures in *continuous* media. (**FBD** 7-1, 7-2).
4. *4th Week*
Review of the methods of chemical kinetics. Homogeneous reactions. (Class Notes)
5. *5th Week*

Polymerization, biochemical and fermentation kinetics. (**FBD** Chapter 1; LA Chapter 9, pp532; **OB** Chapter 7).

6. *6th Week*

Polymerization, biochemical and fermentation kinetics (Cont.) (**FBD** Chapter 1; LA Chapter 9, pp532; **OB** Chapter 7).

7. *7th Week*

Lumping analysis in kinetics, Continuous reaction mixtures. Parameter estimation in linear kinetic systems. The Wei-Prater analysis. (Class Notes)

8. *8th Week*

Rate constants determination for n 1st order reactions (Class Notes)

9. *9th Week*

Parameter estimation in nonlinear kinetic systems. Quasilinearization techniques. Heterogeneous, catalytic reactions. (Class Notes; **FBD** Chapter 2; **JC** Chapter 3).

10. *10th Week*

Review of the isothermal effectiveness factor. Effectiveness factors using Stefan-Maxwell equations. (Class Notes; **FBD** Chapter 3).

11. *11th Week*

Non-isothermal analysis of the catalyst pellet. Stability and dynamics of the catalyst pellet. (Class Notes).

12. *12th Week*

Non-catalytic reactions. Analysis on "moving" reaction domains. (Class Notes; **FBD** Chapter 4). Catalyst poisoning. Reactor-Regenerator combination.

13. *13th Week*

Packed bed reactor modeling. Heterogeneous models for packed bed reactors. (**FBD** Chapter 11).

14. *14th Week*

Modeling and analysis of the fluidized bed reactor. (**FBD** Chapter 13).

15. *15th Week*

Adaptation of conservation equations to various types of reactors. The batch reactor. The continuous, stirred tank and tubular reactors. Reactor models: The perfectly back-mixed, plug-flow tubular and axially dispersed reactor models. (Class Notes; **FBD** Chapters 7-10). Non-ideal continuous flow reactors. The role of residence time distributions. (**FBD** Chapter 12). Turbulent mixing and effect on chemical reactions.

Obviously, the wide variety of topics above cannot be dealt with in uniform detail. Also, modeling of other special reactors such as in chemical vapor deposition is

considered in this course through examples and homework problems since they contain only configurational differences in reaction-diffusion situations. More pointed reading material may be assigned during the course.

Homework Assignments:

Homework assignments will be handed out every two weeks on Mondays and their solutions will be due back on every other Monday.

Grading Scheme:

The course grade will be evaluated based on the performance in homework assignments, mid-semester examination, final examination, and final project presentation. The weightage for each component is given below-

i) Homework Assignments - 20%: There will be a total of 7 homework assignments in this course.

ii) Mid-semester Exam – 25%: This will be open book, and open notes exam.

iii) Computer Exam – 25%: This will be a computer-based exam, which will involve Matlab to solve reaction engineering problems.

iv) Final Project – 30%: This is a team project with no more than 3 students per team. The assessment will be based on technical quality of project report, and presentation.

The minimum points required for different grades are-

Grade A: > 80

Grade B: > 70

Grade C: > 60

Grade D: < 60

Ethics and Academic Dishonesty:

Students in CHE 321 are continuing on a program of study to become chemical engineers. Engineers are expected to uphold a strict code of ethics in which they: (1) build their professional reputations on the merits of their services and (2) act in such a manner as to uphold and enhance the honor, integrity, and dignity of the engineering profession. In addition to these broad ethical guidelines, students in CHE 321 will be expected to be honest and refrain from cheating on exams, and homework assignments (you should not turn in a homework that you did not work yourself).

Students caught cheating will be prosecuted to the maximum extent possible under UIC guidelines. Personally, I take all accusations of academic dishonesty very seriously, and I am asking each you to report any incidents or suspected incidents of

cheating to me directly. I will keep your identity confidential.

Assignments: Any assignment you turn in is to be a product of your own efforts and your own understanding. If you do not understand what you are writing – DON'T WRITE IT. Anyone who copies the solution from another student or a solution manual is guilty of academic dishonesty. Anyone suspected of copying may be given an announced quiz to determine if their work was their own. Even if you "worked with someone", you must do it yourself.